MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 09 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM:** 30

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, every intermediate algebraic step, and boxed final answers with units.
2. In internal choice (**OR**) questions, both options are solved completely with identical CBSE step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Graph scales are maintained uniformly.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. The graph of the linear equation $x = -2$ is a line: [1]

(A) parallel to the x-axis (B) parallel to the y-axis (C) passing through origin (D) perpendicular to y-axis

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) parallel to the y-axis** [0.5 Mark]

Justification: $x = c$ ($c = -2 \neq 0$) keeps abscissa constant at -2 for all y, forming a vertical line **parallel to the y-axis**. [0.5 Mark]

Q2. The graph of the linear equation $y = 5$ is a line: [1]

(A) parallel to the x-axis (B) parallel to the y-axis (C) inclined at 45° to axes (D) passing through origin

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) parallel to the x-axis** [0.5 Mark]

Justification: $y = 5$ keeps ordinate constant at 5 for all x, forming a horizontal line **parallel to the x-axis**. [0.5 Mark]

Q3. The pair of equations $x = a$ and $y = b$ graphically represents lines which are: [1]

(A) parallel (B) intersecting at (b, a) (C) coincident (D) intersecting at (a, b)

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) intersecting at (a, b)** [0.5 Mark]

Justification: Vertical line $x = a$ and horizontal line $y = b$ are perpendicular and intersect uniquely at $(x, y) = (\mathbf{a}, \mathbf{b})$. [0.5 Mark]

Q4. If $2x - 3y = 7$ and $(a+b)x - (a+b-3)y = 4a + b$ represent coincident lines, then a and b satisfy: [1]

(A) $a + 5b = 0$ (B) $5a + b = 0$ (C) $a - 5b = 0$ (D) $5a - b = 0$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: $\boxed{\textbf{(C)}\ a - 5b = 0}$ [0.5 Mark]

Justification: $\frac{2}{a+b} = \frac{3}{a+b-3} = \frac{7}{4a+b}$. From first two: $a + b = -6$. From $\frac{2}{-6} = \frac{7}{4a+b} \implies 4a + b = -21$. Thus $a = -5, b = -1 \implies a - 5b = -5 - 5(-1) = \mathbf{0}$. [0.5 Mark]

SECTION B: Assertion-Reason Questions MATHS MANTHAN | (2 × 1 = 2 Marks)

Q5. **Assertion (A):** The graph of the equation $2x + 3y = 6$ does not pass through the origin $(0, 0)$.
Reason (R): A straight line $ax + by + c = 0$ passes through the origin if and only if $c = 0$. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: $\boxed{\textbf{(A)}}$ [0.5 Mark]

Verification: Here $2x + 3y - 6 = 0$ has $c = -6 \neq 0$ (also $2(0) + 3(0) = 0 \neq 6$). Thus both (A) and (R) are true and (R) explains (A). [0.5 Mark]

Q6. **Assertion (A):** The system of equations $x + y = 8$ and $x - y = 2$ represents intersecting lines.
Reason (R): A pair of linear equations represents intersecting lines if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: $\boxed{\textbf{(A)}}$ [0.5 Mark]

Verification: $\frac{a_1}{a_2} = \frac{1}{1} = 1$ whereas $\frac{b_1}{b_2} = \frac{1}{-1} = -1$, so $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$. Both true and (R) explains (A). [0.5 Mark]

SECTION C: Very Short Answer Questions MATHS MANTHAN | (3 × 2 = 6 Marks)

Q7. If $x = a$ and $y = b$ is the solution of the equations $x - y = 2$ and $x + y = 4$, find the values of a and b. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Substitution & Elimination [2 Marks]

Substituting (a, b): $a - b = 2$ — (i) and $a + b = 4$ — (ii).
Adding (i) and (ii): $2a = 6 \implies a = 3$. From (ii): $3 + b = 4 \implies b = 1$. $\boxed{\mathbf{a = 3,\quad b = 1}}$

Q8. Check whether the lines $x - 2y = 0$ and $3x + 4y - 20 = 0$ intersect at a unique point, and find the point algebraically. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Uniqueness Check [1 Mark]

$\frac{a_1}{a_2} = \frac{1}{3} \neq \frac{b_1}{b_2} = \frac{-2}{4} = -\frac{1}{2} \implies$ the lines intersect at a unique point.

Step 2: Finding the Intersection Point [1 Mark]

From $x - 2y = 0 \implies x = 2y$. Substituting into $3x + 4y - 20 = 0$: $3(2y) + 4y = 20 \implies 10y = 20 \implies y = 2 \implies x = 4$.

$\boxed{\textbf{Yes, unique intersection at } (4, 2)}$

Q9. Find the value of k for which the system of equations $x - ky = 2$ and $3x + 2y = -5$ has a unique solution. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1 & 2: Ratio Condition [2 Marks]

For a unique solution, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \implies \frac{1}{3} \neq \frac{-k}{2} \implies -3k \neq 2 \implies \boxed{\mathbf{k \neq -\frac{2}{3}}}$.

Q10. In a $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$. Find the measures of all three angles. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let $\angle A = x^\circ$ and $\angle B = y^\circ$, so $\angle C = 3y^\circ$.

Formula / Theorem Used: Angle Sum Property of a Triangle: $\angle A + \angle B + \angle C = 180^\circ$

Step 1: Framing Two Equations in x and y [1.5 Marks]

From $3\angle B = 2(\angle A + \angle B)$: $3y = 2(x + y) \implies 2x - y = 0 \implies y = 2x$ — (i)

From $\angle A + \angle B + \angle C = 180^\circ$: $x + y + 3y = 180 \implies x + 4y = 180$ — (ii)

Step 2: Solving for $\angle A$, $\angle B$, $\angle C$ [1.5 Marks]

Substituting $y = 2x$ into (ii): $x + 4(2x) = 180 \implies 9x = 180 \implies x = 20^\circ$.

Thus $\angle A = 20^\circ$, $\angle B = 2(20^\circ) = 40^\circ$, and $\angle C = 3(40^\circ) = 120^\circ$.

$$\boxed{\angle A = \mathbf{20}^\circ, \quad \angle B = \mathbf{40}^\circ, \quad \angle C = \mathbf{120}^\circ}$$

Q11. Find the values of a and b for which the following system of linear equations has infinitely many solutions:

$$2x + 3y = 7 \quad \text{and} \quad (a - b)x + (a + b)y = 3a + b - 2$$

[3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Equating Ratios [1.5 Marks]

$$\frac{2}{a-b} = \frac{3}{a+b} = \frac{7}{3a+b-2}$$

From first pair: $2(a + b) = 3(a - b) \implies 2a + 2b = 3a - 3b \implies a = 5b$ — (i)

From last pair: $3(3a + b - 2) = 7(a + b) \implies 9a + 3b - 6 = 7a + 7b \implies 2a - 4b = 6 \implies a - 2b = 3$ — (ii)

Step 2: Solving for b and a [1.5 Marks]

Substituting $a = 5b$ into (ii): $5b - 2b = 3 \implies 3b = 3 \implies b = 1$. Thus $a = 5(1) = 5$.

$$\boxed{\mathbf{a = 5}, \quad \mathbf{b = 1}}$$

Q12. Draw the graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the y-axis, and calculate its area. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Tables of Values [1 Mark]

x	0	2	-1
$y = x + 1$	1	3	0

x	0	2	4
$y = \frac{12-3x}{2}$	6	3	0

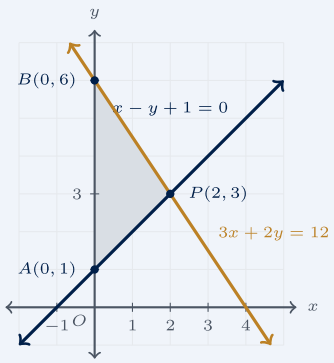
Step 2: Vertices with y-axis [1 Mark]

Lines intersect at $P(2, 3)$ and cut the y-axis ($x = 0$) at $A(0, 1)$ and $B(0, 6)$.

Step 3: Area of $\triangle PAB$ [1 Mark]

Base AB along y-axis $= 6 - 1 = 5$ units, Height $= x_P = 2$ units.

$$\text{Area} = \frac{1}{2} \times 5 \times 2 = \mathbf{5} \text{ sq. units}$$



$$\boxed{\text{Vertices: } \mathbf{(0,1)},\ \mathbf{(0,6)},\ \mathbf{(2,3)}; \quad \text{Area} = \mathbf{5} \text{ sq. units}}$$

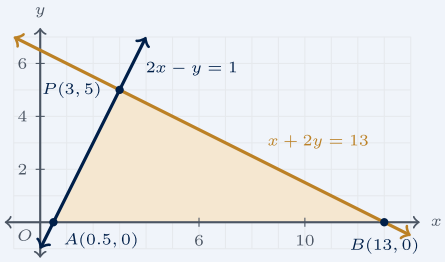
Q13. Draw the graphs of the linear equations $2x - y = 1$ and $x + 2y = 13$. Find the coordinates of the point of intersection of the lines, the coordinates of the vertices of the triangle formed by these lines and the line $y = 0$, and calculate the area of this triangle. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Solution Tables [1.5 Marks]

x	0.5	1	3
$y = 2x - 1$	0	1	5

x	13	5	3
$y = \frac{13-x}{2}$	0	4	5



Step 2: Intersection & Vertices [1.5 Marks]
Both lines meet at $P(3, 5)$ and intersect the x-axis ($y = 0$) at $A(0.5, 0)$ and $B(13, 0)$.

Step 3: Area Calculation [2 Marks]
Base $AB = 13 - 0.5 = 12.5$ units, Height $= y_P = 5$ units.

$$\text{Area} = \frac{1}{2} \times 12.5 \times 5 = \frac{62.5}{2} = \mathbf{31.25} \text{ sq. units}$$

Intersection: $(\mathbf{3}, \mathbf{5})$; Vertices: $(\mathbf{0.5}, \mathbf{0})$, $(\mathbf{13}, \mathbf{0})$, $(\mathbf{3}, \mathbf{5})$; Area = **31.25** sq. u.

Q14. CASE STUDY: Sports Academy Equipment Procurement

The sports department of a school is preparing for an inter-school tournament. The head coach placed two consignments with an equipment dealer: First consignment: 7 cricket bats and 6 leather balls for Rs. 3800. Second consignment: 3 cricket bats and 5 leather balls of the same quality for Rs. 1750. Let the price of one bat be Rs. x and the price of one ball be Rs. y. [4]

(i) Represent the two purchases algebraically as a pair of linear equations in x and y. [1]

(ii) Find the unit price of a single cricket bat (x). [1]

(iii) Find the unit price of a single cricket ball (y). [1]

(iv) What would be the total bill if the coach places an order for 5 bats and 4 balls? [1]

— OR —

Find the difference between the price of 2 bats and 10 balls. [1]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Sub-part (i): Algebraic Equations [1 Mark]

$7x + 6y = 3800$ — (i) and $3x + 5y = 1750$ — (ii)

Sub-part (ii): Unit Price of One Cricket Bat (x) [1 Mark]

Multiplying (i) by 5 and (ii) by 6: $35x + 30y = 19000$ and $18x + 30y = 10500$.

Subtracting: $17x = 8500 \implies \boxed{\mathbf{x = \text{Rs. } 500}}$.

Sub-part (iii): Unit Price of One Cricket Ball (y) [1 Mark]

Substituting $x = 500$ in (ii): $3(500) + 5y = 1750 \implies 5y = 250 \implies \boxed{\mathbf{y = \text{Rs. } 50}}$.

Sub-part (iv): Total Bill for 5 Bats and 4 Balls [1 Mark]

Bill $= 5(500) + 4(50) = 2500 + 200 = \boxed{\text{Rs. } \mathbf{2{,}700}}$.

Sub-part (iv) [OR Option]: Difference Between 2 Bats & 10 Balls [1 Mark]

Difference $= 2(500) - 10(50) = 1000 - 500 = \boxed{\text{Rs. } \mathbf{500}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★